Bangladesh University of Engineering and Technology
Department of Computer Science and Engineering

Course: CSE 206
Digital Logic Design Sessional

Experiment No. 4
Arithmetic circuit design.

1. Design a 1-bit full Subtractor circuit using logic gates.
2. Using basic gates, design a 2-bit comparator to compare two 2-bit numbers X and Y. The circuit should provide 3 output lines to indicate $X > Y$, $X = Y$, and $X < Y$.

Lab Sheet: For each of the problems/questions, your report should cover the following points:

- Problem specification.
- Required IC chips.
- Truth table.
- Simplification steps.
- Required equation in minimized form.
- Circuit diagram with pin number.

Answer the following question.

1. Design a 4-bit Carry Look Ahead (**CLA**) adder circuit.
2. Add an overflow output to your adder circuit. The overflow output should be on whenever the output is too large to be represented by 4 bits.
3. Design a 4-bit BCD (8-4-2-1) adder.